

MADE IN INDIA



UNIK SURGICAL PVT. LTD.

Manufacturer, Importer, Exporter & Suppliers

Medical Furniture, Lab Furniture, Non Medical
Furniture, MOT, MGPS, Laundry Equipments, Kitchen Equipments, CSSD, TSSU,
BMW, Medical Equipments, Mortuary Equipments, Nurse Call System,
Lab Equipments & General Orders Supplier



MODULAR OPERATION THEATRE



QUALITY | TECHNOLOGY
COMMITMENT | SERVICES



UNIK SURGICAL PVT. LTD.

Founded in 1999, Unik Surgical Pvt. Ltd. has been at the forefront of healthcare infrastructure development in Eastern India. What began as a small enterprise has grown into a leading supplier and manufacturer of medical and surgical equipment and furniture.

With our headquarters in Patna, Bihar, a regional office in Greater Noida, Uttar Pradesh, and a manufacturing facility in Chapra, Bihar, we serve clients across the region with unmatched expertise and a strong commitment to quality. Our journey is defined by continuous innovation, strict adherence to international standards, and an unwavering dedication to improving healthcare facilities across India.

Contracting

Project Report

Layout

Configuration

Consulting

Installation

On-site Training

Startup

Costing

Technical Support

ABOUT DIRECTOR

Manoj Kumar Singh is a healthcare industry leader known for his practical approach and deep commitment to improving patient care. With years of hands-on experience in medical equipment manufacturing, he founded Unik Surgical with a simple goal — to make reliable, high-quality surgical solutions accessible to healthcare facilities of all sizes.

What began as a trader in Patna has now grown into a trusted name, serving hospitals and clinics across the country. Manoj's focus on precision engineering, transparent business ethics, and long-term client relationships has helped Unik Surgical earn the confidence of medical professionals.

His leadership continues to guide the company toward innovation and responsible growth, while staying rooted in the belief that every piece of equipment should ultimately contribute to better patient outcomes.



International & National Accreditation of Bodies



एन. एस. आई. सी.
NSIC
ISO 9001 : 2008

CERTIFICATIONS



QUALITY | TECHNOLOGY
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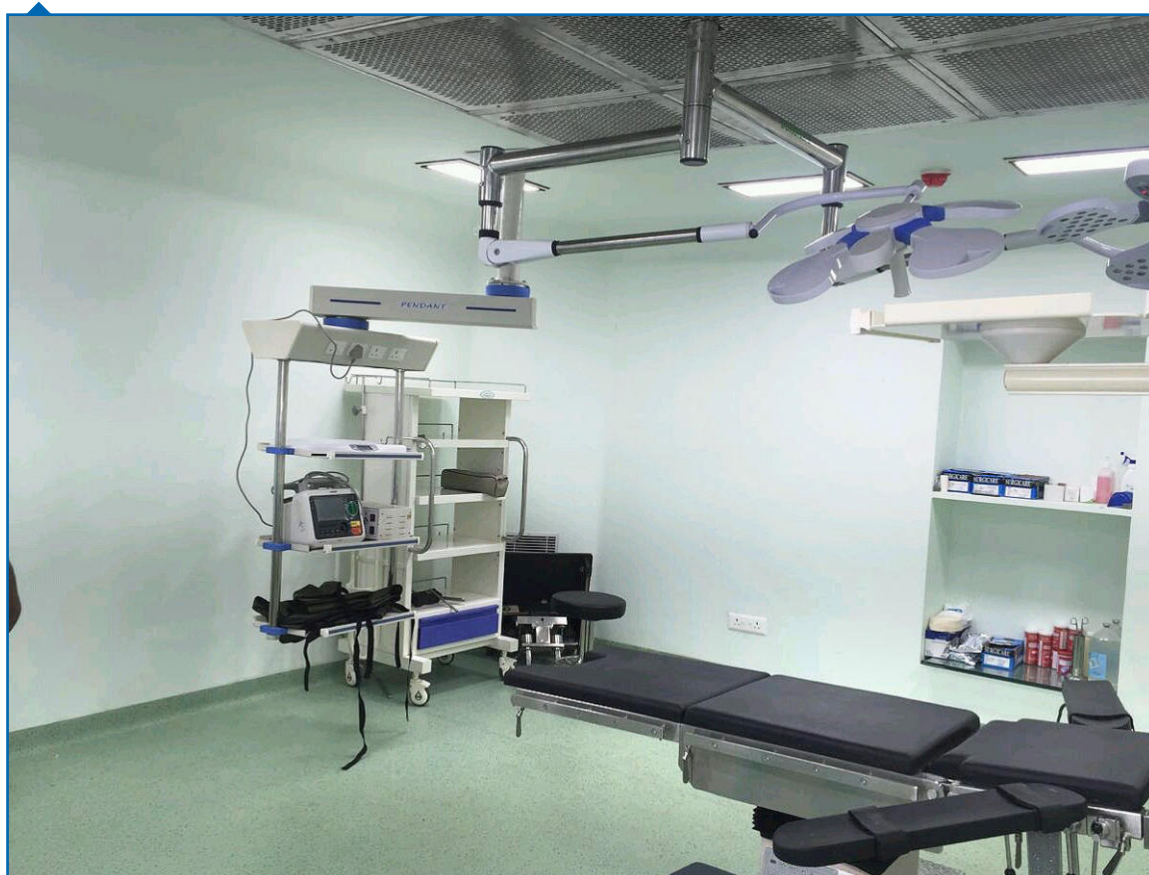
PRODUCTS MANUFACTURED BY US

- Pre-Fabricated modular Operation Theatres in Galvanised Steel Panels, High Pressure Laminate Panels, PUF Panels as per Customer requirement.
- Ceilling Mounted Laminar Flow (Planair) Systems
- OT Control Panels
- X-ray View Panels
- Operating Magnetic List Boards
- Storage Units
- Clean Room General Lights, OT Lights
- OT Floor (Antistatic Conductive Tiles)
- Hermetically Sealed Sliding Door (Manual / Automatic)
- Stainless Steel Clean Room Doors
- Bed Head Panels
- Perssure Relief Dampers
- Multi Movement Pendant Systems •Nurse Call Systems
- Automatic Surgical Scrub Stations
- Sealed Swing Doors
- Hatch Boxes
- ICUs Cubical Partition Curtains
- Medical Gas Pipeline Systems & Related Products



WALL & CEILING CONSTRUCTION

The wall and ceiling panel can be made out of following material as per customer requirement, a) 1.6mm thick EGP steel panels backed by 12mm thick Gypsum, b) Stainless steel SS304 panels, c) High-pressure Laminate panels as per EN standard, d) PUF sandwiched panel. The room wall will have two independent surfaces with minimum opening in between. The external walls of the room will be constructed with solid bricks with cements Plastering. The inner surfaces walls will be constructed with 1.60 mm thick EGP steel panels backed by 12-mm gypsum board (India gypsum make) these panels will have flame resistance to Bs 1142 part 3. The inner surface walls will be fixed to the bricks with Essential supports. There will be minimum possible cavity/gap in between the solid and steel walls. The total distance between the inside and outside surfaces of the operating room will be variable to suit the architect's layout, but will be sufficient for the flush mounting of equipments. 2400mm X 2800mm, 2800mm X 2800mm, as per theatre size.



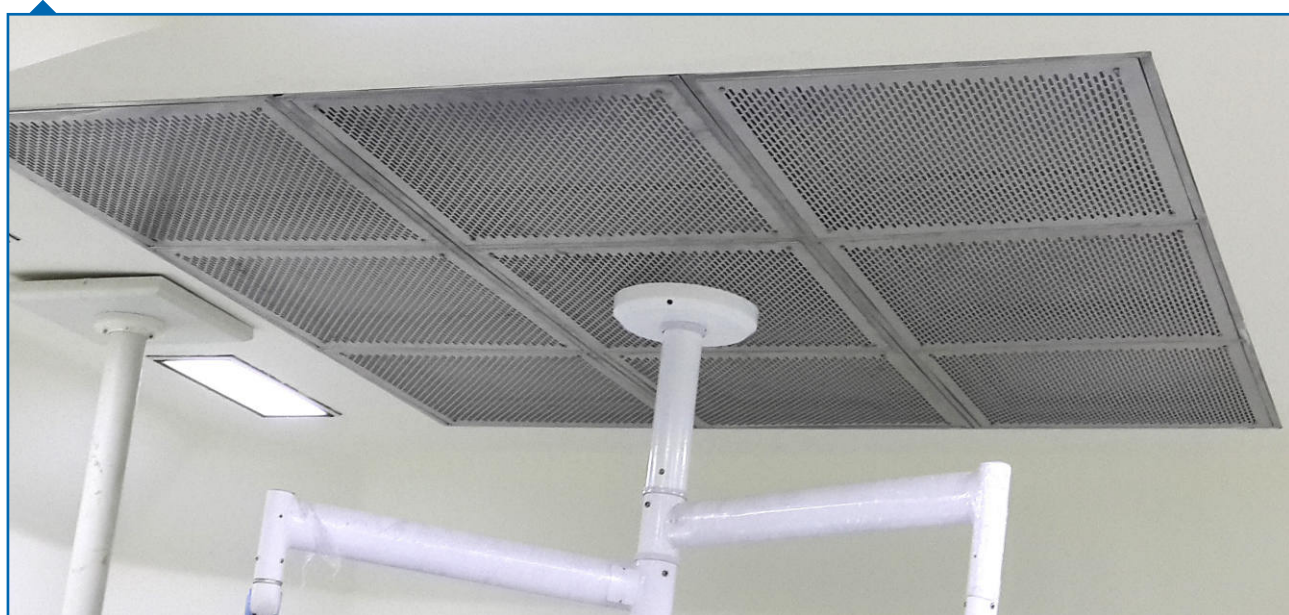
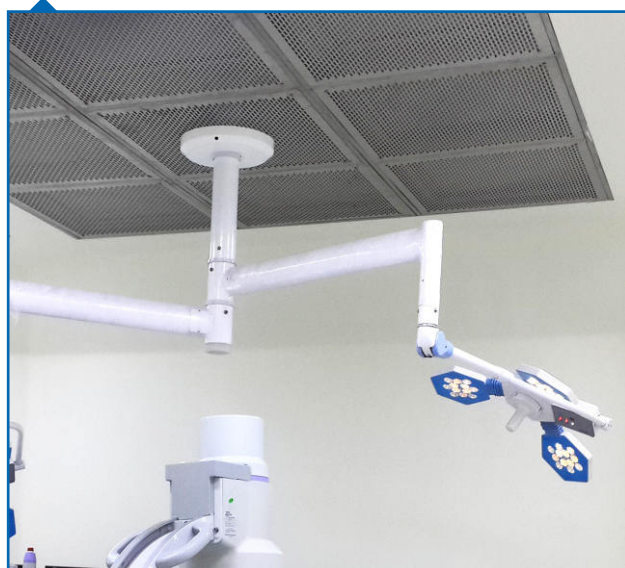
AIR MANAGEMENT SYSTEM (PLENUM) (JK MAKE)

The laminarflow system shall comprise of the thick extruded aluminum profiles frame. The complete unit will have factory prepared fine sealing system along with proper test certificates. The plenum will be supplied at silt duty sealed in factory made packing. The Corners of the plenum will be made of extruded aluminum sections which support the fire retardant FRP housings in such a manner that the air is passed only through the deep pleat HEPAfilter only. The laminar flow system will have SS perforated diffuser grill. The laminarflow system will have such design that it provides cleanliness of class 100.(≤ 100 particles/ft³) and bacteriological class B (≤ 20 cfu/m³).

Perfect tightness guaranteed by a seal between filters and holding structure enabling no by pass of HEPA Filters. Complete Filtration System will be factory assembled, tested and packed.

The absolute filters installed in the plenum will be suitable for applications for Laminarflow and clean rooms, these absolute filters will be deep pleat HEPA filters having extruded anodized aluminum,69 mm deep frame, and filter will provide flow uniformity exceeding+-

1 Protective grids	White epoxy painted micro drawn grid
2 Separators	Continuous thermo plastic chord
3 Sealant	Polyurethane
4 Gasket	One piece polyurethane
5 EN 1822 class	H14
6 MPPS average efficiency	> _ 99.995%
7 MPPS average efficiency	> . gg 975%
8 3 micron DOP efficiency	> _ gg 999%
9 Final pressure drop	60Q p'a (maximum)
10 Maximum operating Temperature	60 degree centigrade
11 Maximum RH	90 percent
12 Efficiency Tests	Filters individually tested and certified



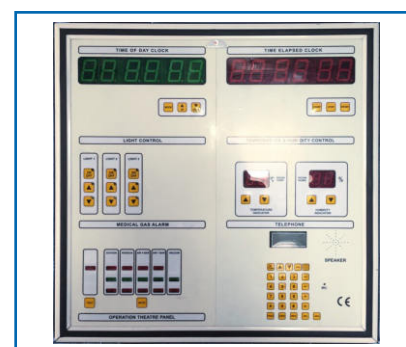
SURGEON CONTROL PANEL

The surgeon control panel should meet electrical safety codes for high & low voltage system, wired to the current IEE regulations. The surgeon's control panel should be designed to cope with changing technology & equipment in operating environments. Control panel will be user friendly & ease of operating & maintaining purpose. The panel should be "Membrane" type, configured room staff required. The fascia should be made superior quality UV resistance membrane with sterilization feature.

The Panel should contain 6 or 9 service tiles below:

1. Time Day Clock
2. Time Elapse Day Clock
3. General Lighting System
4. Medical Gas Alarm Panel
5. Hand Free Telephone set
6. Music control
7. Temperature & Humidity Display and or control

- Time days clock be digital type & clocks having high brightness characters.
- Temperature indicator should indicate the room temperature which should be connected to the local pressure switches of Air-Conditioning System.
- Indicators should be digital type & clock having high brightness character, not less than 30 mm in height.
- Central lighting system should incorporate all the necessary control of the lighting system inside the theater.
- The medical gas alarm should indicate high, Normal & Low gas Pressure for each gas service present in the operating system & should have an audible buzzer with mute facility. Pressure sensors should be connected to MGPS for monitoring the pressure.
- A hand free set Telephone system should be incorporated in the panel with memory type card.



TOUCH SCREEN SURGEON
CONTROL PANEL

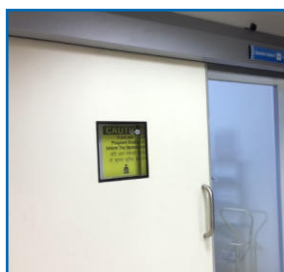


HERMETICALLY SEALED SLIDING DOORS

To maintain satirically and the correct air pressure the room, all doors into and out should be of the sliding, hermetically sealing type. The doors should meet following specifications. Meets international quality & safety requirements.

- Doors should be wired to the current IEE regulations & Bs7971 standard.
- Motor should be DC 24V 70W brush less DC motor.
- Noise level of movement should not be more then 60 decibel.
- controller should able microprocessor based and the be CE market.
- Power efficiency should be .95 (in AC 100V full load).
- The track should be made up of single piece extruded aluminum.
- Environment temperature should be - 200C to +550C.
- Starting time should be able to regulate from .5 second to 23 second & starting speed should be 600 mm persecond.
- Electrical safety codes for high & low voltage system.

Design should meet HTM 2020/2021 standards. The doorframe should be made of high quality anodize aluminum and the door panel should be made of compact laminate that can withstand high abrasion. To ensure efficient sealing of the doors frames should be provided. They will consist of reinforced plasterboard panels faced with same laminate as the door should seal on all four edges in the closed position & should be surface installed type. The track of the door should be constructed with high quality door lock with aluminum extrusion, fixed firmly to the door in such a way they do not obstruct trolley movement through the door. The door frames should be edged with an aluminum extrusion & with concealed fixings that are adjustable during installation to ensure a 100% hermetic seal is achieved. Vision panels, 300mm X 300mm should be provided in the doors. The door controller should be sensing overload condition and in overload case the door will automatically stop & reverse the direction of travel. The controller should be capable of either being operated by elbow switches/foot switched, radar switch(touch less sensor). All doors should be able to be operated easily manually in the event of failure of the power supply of the automation unit.



X-RAY VIEW PANEL

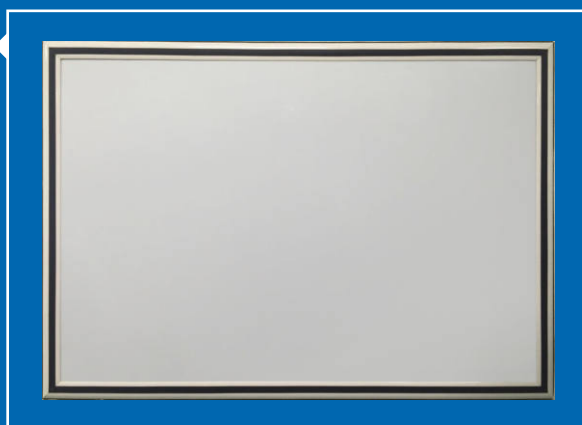
JK Engineering works provides a range of x-ray viewing screens (any size) meet the electrical safety codes for high and low voltage system. The theatre is to be equipped with a 2 plate X-ray viewing screen, incorporating high efficiency electronic ballasts, and electronic digital dimming control (101 % to 10 %) circuit to provide flicker free luminance for the film viewing purpose. It can be installed flushed with theatre wall for hygienic and ease of cleaning purpose or desktop model. The X-ray viewing screen is designed for the purpose of front access. JK Engineering x-ray viewing screens provide high levels of luminous density from 4000 to 5400 cd/sqm. JK engineering range of X-ray viewing screens is ideally suited for general, surgical and dental using modern hospitals. The flush and wall mounted models are supplied with 2 meters of heat resistant cable, with plug, available in one, two, three, four, six and nine plate models. The glare free diffuser has spring loaded white clips to secure the x-ray.



MAGNETIC BOARD

Operating List Board

One operating list board will be provided in each operating theater. It will be made of ceramic having Magnetic properties and will be flushed to the wall of the operating room. It can be any size as per requirement of.



PRESSURE RELIEF DAMPER

- The cascade pressure stabilisers are a range of multi bladed units specifically designed to control room air pressures in critical areas such as Operation theatres etc.
- Each Stabiliser comprises of a carbon steel case & matching slip over ring.
- The carbon steel housing contains up to four Grade 304 stainless steel Blades, which pivot upon sealed for life bearing assemblies.
- Balancing should be carried out utilising a proven balance weight assembly. Adjustment can easily be carried out on site should the need arise. Structural steel frames All structural steel sections will be of Grade 43 to BS 4360 or equivalent Indian Standard.
- The structural frames for the operating theatre are designed taking into account all fixed equipment to be installed in the Modular OT.
- The theatre structure should support all equipment installed in the Modular OT, such as the operating lamp etc, and should be vibration free and rigid.
- The theatre structure should also be capable of supporting other equipment to be installed in the same Modular OT in future, with total loading not more than twice that of all the equipment to be installed in the OT as mentioned in this specification. Welded sections in accordance with BS 5135.



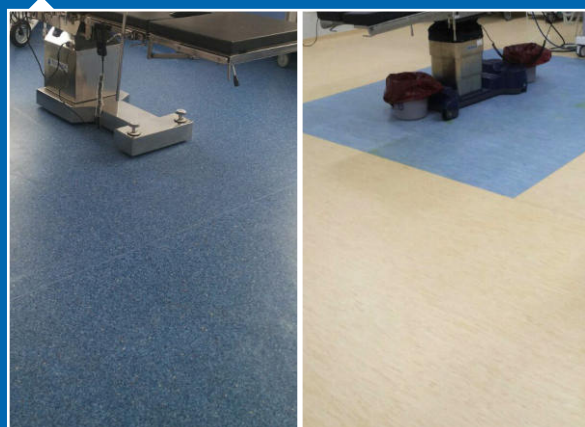
HATCH BOX (PASS BOX)

A Hatch of 600mm x 600mm size will be provided in the Operation Theatre as specified in the scope of the work to remove waste materials from the Operation Theatre to Dirty linen Area just adjacent to Operation Theatre. Each Hatch will be equipped with two doors and the door will be operated electronically. The Hatch will be designed in such a way that only one door will be opened at one time. The UV light will be so installed that it is kept on while both the doors are closed, this UV light has to be automatically turned off in case of opening of either of the doors. There will be indicators on both sides of the OT so that the door open/close status can be monitored from both ends.



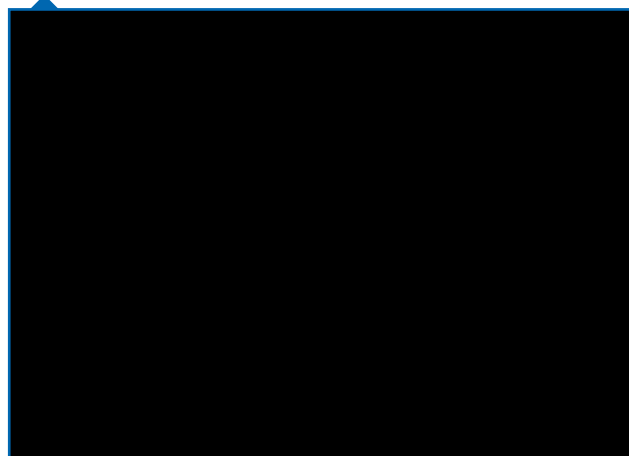
OT FLOORING (ANTISTATIC CONDUCTIVE TILES/ROLLS)

A floor screed Should be provided, flat to within a tolerance of +/-3mm over any 3 metre area. Onto this sub-floor, a self-leveling compound should be laid prior to laying of the floor finish. Copper grounding strips (0.05m thick, 50 mm width) should be laid flat on the floor in the conductive adhesive and connect to copper wire of grounding. The floor finish in the operating room should be 2mm Conductive PVC tiles, laid on semi-conductive adhesive base. The floor finish should terminate at the room perimeter passing over a concealed cove former and continuing up the wall for 100mm. All joints should be welded with electrodes of the same compatible material to provide a continuous sealed surface. The floor should have an electrical resistance of 2.5×10^6 to 10^6 Ohms, as per DIN 51953 ATM F-1450 or NFPA 99, B1 class of fire resistance and should meet UL standard 779. Fulfills product requirements as per EN 649. To maintain sterility and the correct air pressure in the room all doors into an OT should be of the sliding, hermetically sealing type. The door should meet following specifications.

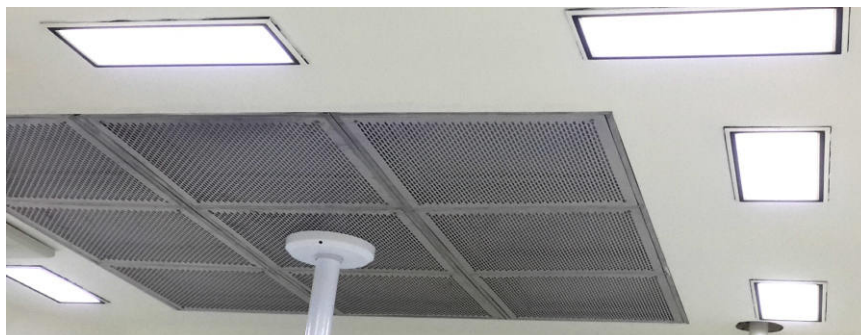


VIEW WINDOWS WITH BLINDS

The view window of specified size shall be providing consisting of: Double insulated fixed glazing with not less than 5mm thick toughened glass. Window frame shall be powder coated Aluminum of approved shape flush mounted with wall paneling. Motorized horizontal Venetian Blinds of powder coated aluminum strips of vista level of equivalent of approved shade including necessary accessories. The motor shall be of reputed brand approved by Engineer-in-charge. The Venetian blinds should be motorized for 90 degree rotational.



PERIPHERAL LIGHTS (LED/CFL)



STORAGE UNIT

The storage unit can be made with 1.50 mm thick EGP Zinc coated steel panels, Stainless Steel panels, High pressure Laminate panels. The storage unit will be divided in 2 equal parts and each part will have individual glass doors with high quality locking system. Each part will be provide with glass racks.



SURGICAL SCRUB SINK

- Compact Surgical scrub Sink should be designed for use in Operation theatre complex providing surgeons with a convenient sink for pre-op scrub up.
- Each Fixture should be fabricated from heavy gauge type 304 stainless steel & should be seamless welded construction polished to a satin finish.
- The scrub sink should be provided with a front access panel, which should be easily removed for access to the water control valve, waste connections, stoppers & strainers.
- Hands free Operation includes infrared sensor with built-in range of adjustment.
- Thermostatic Mixing Valve control should be located behind the access panel & maintain constant water temperature.
- User defined settings of 1,3,5 & 10 Min. Should be available. This timing can be changed to meet individual application requirement. Provided with elbow action taps, infrared sensor thermostatic control with fail-safe temperature controls.
- All units should have radiused anti splash fronts. Knee/ foot operated switch should be offered as an option.





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